WORKED SOLUTION

STEP II 2006, Question 9

Mechanics • Friction and Toppling

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Read the question before you go on. It is in the STEP Support Programme's question database:

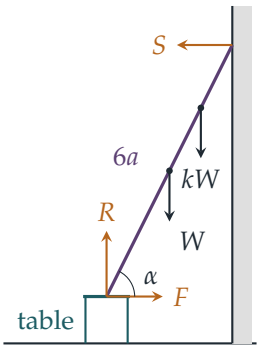
step.maths.org/questions/06-s2-q9

The key idea

Treat the ladder and the table as two separate bodies, linked by the contact at the foot of the ladder. The ladder alone tells us the reaction and the friction at its foot, and these depend on how high the painter is. Then Newton's third law hands the same forces, reversed, to the table. For the table we compare two thresholds: the push that makes it slide, and the push whose moment about the far legs beats the moment of the weight on it.

Solution

We model the ladder as a uniform rod, the painter as a particle, the wall as smooth and the table as a rigid body with its top at height a (we ignore its thickness). Write $\alpha = \arctan 2$, so $\tan\alpha = 2$, $\sin\alpha = \frac{2}{\sqrt{5}}$ and $\cos\alpha = \frac{1}{\sqrt{5}}$. Let the painter stand a distance x along the ladder from its foot, where $0 \le x \le 6a$.



Part (i)

The forces on the ladder are: its weight W at its midpoint, the painter's weight kW, the horizontal reaction S from the smooth wall, and at the foot a normal reaction R

upwards and a friction force F towards the wall. (Friction must point towards the wall, since it is the only horizontal force that can balance S.)

Resolving vertically and horizontally,

$$R = (k+1)W, \qquad F = S.$$

Taking moments about the foot of the ladder, the top is at height $6a\sin\alpha$ above the foot and the weights act at horizontal distances $3a\cos\alpha$ and $x\cos\alpha$, so

$$S \cdot 6a\sin\alpha = W \cdot 3a\cos\alpha + kW \cdot x\cos\alpha.$$

Dividing by $6a\sin\alpha$ and using $\cot\alpha = \frac{1}{2}$,

$$F = S = \frac{W}{12}\Big(3 + \frac{kx}{a}\Big).$$

This increases with x, so the largest friction is needed when the painter is at the very top, $x = 6a$:

$$F_{\max} = \frac{(3+6k)W}{12} = \frac{(2k+1)W}{4}.$$

The greatest friction available is

$$\tfrac{1}{2}R = \frac{(k+1)W}{2} = \frac{(2k+2)W}{4}.$$

Since $2k+1 < 2k+2$, we have $F \le F_{\max} < \frac{1}{2}R$ for every position of the painter, and for every k. With the table fixed, the ladder therefore never slips, however high the painter stands, as required.

Part (ii)

Now $k = 9$, so $R = 10W$ and

$$F = \frac{W}{4}\Big(1 + \frac{3x}{a}\Big),$$

which rises from $\frac{1}{4}W$ at $x = 0$ to $\frac{19}{4}W$ at $x = 6a$. Part (i) still applies, so the ladder does not slip on the table top.

By Newton's third law the ladder pushes on the table top, at its centre, with a force $R = 10W$ downwards and a horizontal force F *away from* the wall. The table also has its own weight $2W$ acting at its centre, and the ground acts on its legs with normal reactions and friction.

Ask yourself How can a horizontal push at the centre of the table top make the table tilt?

The push F acts at height a, so it has a moment about the bottom of the two legs furthest from the wall. If the table tilts, it rotates about the line joining the feet of those legs. The legs are at the corners of a square of side $\frac{1}{2}a$, so that line is a horizontal distance $\frac{1}{4}a$ from the centre of the table.

Tilting. About the line of the far legs' feet, the ground's forces on the far legs have no moment, and nor does ground friction on any leg (it acts at ground level). The downward forces $10W + 2W = 12W$ at the centre have a restoring moment $12W \cdot \frac{1}{4}a = 3Wa$. The push F has an overturning moment Fa. Taking moments about this line, the normal reaction N_1 (in total) on the two legs nearest the wall satisfies

$$N_1 \cdot \tfrac{1}{2}a = 3Wa - Fa,$$

so $N_1 \geq 0$ exactly when $F \leq 3W$. The table is about to tilt when $N_1 = 0$, that is when

$$\frac{W}{4}\left(1 + \frac{3x}{a}\right) = 3W \quad \Longleftrightarrow \quad x = \frac{11a}{3}.$$

Slipping. Resolving vertically for the table, the total normal reaction from the ground is $12W$, so the greatest total friction available is $\frac{1}{3} \cdot 12W = 4W$. Resolving horizontally, the friction needed is F. So the table is about to slip when

$$\frac{W}{4}\left(1 + \frac{3x}{a}\right) = 4W \quad \Longleftrightarrow \quad x = 5a.$$

Both values lie within the ladder's length $6a$, and F increases steadily as the painter climbs. It reaches $3W$ (at $x = \frac{11}{3}a$) before it reaches $4W$ (at $x = 5a$). So **the table tilts first**. This happens when the painter is $\frac{11}{3}a$ along the ladder, a height $\frac{11}{3}a \sin\alpha = \frac{22a}{3\sqrt{5}} \approx 3.28a$ above the table top. At that moment the friction needed at the ground is only $3W$, less than the $4W$ available, so the table has not slipped.

Checks

- With no painter on the ladder ($x = 0$), $F = \frac{1}{4}W$, and the near legs carry $N_1 = \frac{11}{2}W$ while the far legs carry $\frac{13}{2}W$. The far legs take slightly more, as they should with the push directed away from the wall.
- A computer computes the forces for many values of k and x. In (i) the largest ratio of friction needed to friction available is just under 1 (it is $\frac{2k+1}{2k+2}$ at the top). In (ii) it finds the near-leg reaction vanishing at $x \approx 3.667a$ and the ground friction reaching its limit at $x = 5a$.

Common slips

- Taking the tilting axis at the table's edge a distance $\frac{1}{2}a$ from the centre: the table top has side $\frac{1}{2}a$, so the half-width is $\frac{1}{4}a$.
- Forgetting that the push on the table acts at height a, the top of the legs, not at ground level. It is this height that gives the push its overturning moment.
- Leaving out the ladder and painter's $10W$ when finding the ground's normal reaction on the table. Only the table's own weight $2W$ is not enough.

Beyond STEP

This question is about how a system of rigid bodies can fail in competing ways: by sliding, or by toppling. The method is to find the threshold for each and see which the load reaches first. Engineers apply the same idea to retaining walls, cranes and stacked loads, and in statics courses it leads on to the principle of virtual work. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body equilibrium and friction with care.

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